RUN LOCALLY

1. Introduction to Running Locally

Name: Malathi. K

Running a project locally means starting and testing the application on your own computer instead of deploying it to a public server.

Local development provides a safe environment for building features, checking errors, and making changes before publishing the project.

Key Points

• Local computer
• Source code
• Development server
• Browser testing
• Local database
• Project files

This 10-page report explains the basic process of setting up, running, testing, and debugging a web application locally.

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2. Setting Up the Development Environment

Name: Malathi. K

A development environment contains the software and tools required to create and run an application.

The required setup depends on the project and may include a code editor, runtime environment, package manager, database, and browser.

Key Points

• Code editor
• Runtime installation
• Package manager
• Browser
• Terminal
• Project folder

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3. Project Structure

Name: Malathi. K

A clear project structure makes files easier to find and maintain. Web projects commonly separate source files, assets, configuration, and dependencies.

Organizing files consistently also helps teams understand how different parts of the application are connected.

Key Points

• Source directory
• Public assets
• Configuration files
• Dependencies
• Documentation
• Environment files

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4. Installing Dependencies

Name: Malathi. K

Modern applications often use external libraries and packages. A package manager downloads the required dependencies and records their versions.

Installing dependencies correctly is important because the application may not run if required packages are missing or incompatible.

Key Points

• Package manager
• Dependency list
• Version control
• Installation command
• Lock file
• Compatibility

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5. Starting a Local Server

Name: Malathi. K

A local development server runs the application on the computer and makes it available through a local address in a web browser.

The server may automatically reload or rebuild the application when source files are changed.

Key Points

• Start command
• Local host
• Port number
• Server logs
• Auto reload
• Stop and restart

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6. Environment Variables and Configuration

Name: Malathi. K

Applications often need configuration values such as API endpoints, database settings, or development options.

Environment variables can keep configuration separate from source code and help different environments use different settings.

Key Points

• Configuration
• Environment variables
• API URL
• Development mode
• Local settings
• Secure secrets

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7. Testing the Application Locally

Name: Malathi. K

Local testing helps developers find errors before users access the application. Test pages, forms, navigation, APIs, and important user interactions.

Browser developer tools can help inspect errors, network requests, console messages, and page behavior.

Key Points

• Functional testing
• Browser testing
• Console checks
• Network inspection
• Form testing
• Error checking

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8. Debugging Common Problems

Name: Malathi. K

Local applications can fail because of missing dependencies, incorrect ports, syntax errors, configuration problems, or unavailable services.

Read error messages carefully, check logs, confirm configuration values, and test one problem at a time.

Key Points

• Missing package
• Port conflict
• Syntax error
• Wrong configuration
• Server error
• Connection problem

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9. Local Database and API Testing

Name: Malathi. K

Some applications need a local database or a connection to a development API. This allows developers to test data operations without changing production data.

Database migrations, sample data, and API test tools can support reliable local development.

Key Points

• Local database
• Sample data
• API requests
• CRUD operations
• Database migration
• Test environment

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10. Best Practices and Conclusion

Name: Malathi. K

Running locally is an important part of the software development process. It allows developers to build, test, debug, and improve applications before deployment.

A good local workflow uses organized files, reproducible dependencies, clear configuration, regular testing, and useful error handling.

Key Points

• Keep dependencies updated
• Document setup steps
• Use safe local data
• Check errors early
• Test before deployment
• Keep the project organized